# Sloshing

PENG

October 20, 2026

## Contents

**1 Introduction** **1**

**2 Analytical Method** **1**

## 1 Introduction

**Definition 1.1** (dispersion)**.**

- *frequency dispersion*
- *amplitude dispersion*

**Definition 1.2** (nonlinear resonance)**.** *Nonlinear resonance is a mechanism by which energy is continuously exchanged between a small number of linear wave modes [1].*

## 2 Analytical Method

**Assumption 2.1.** *Potential flow:*

- *irrotational*
- *inviscid*
- *incompressible*

**Definition 2.2.** *Governing equations:*

$$\Delta\phi + \phi_{zz} = 0 \quad \text{for } x \in D, \quad -h < z < \epsilon\eta \tag{2.1}$$

$$\phi_t + \eta + \frac{\epsilon}{2}\left(|\nabla\phi|^2 + \phi_z^2\right) = 0 \quad \text{for } x \in D, \quad z = \epsilon\eta \tag{2.2}$$

$$\eta_t + \epsilon\nabla\phi\cdot\nabla\eta = \phi_z \quad \text{for } x \in D, \quad z = \epsilon\eta \tag{2.3}$$

$$n\cdot\nabla\phi = 0 \quad \text{for } x \in \partial D, \quad -h < z < \epsilon\eta \tag{2.4}$$

$$\phi_z = 0 \quad \text{for } x \in D, \quad z = -h \tag{2.5}$$

*Why does heave excitation lead to either symmetrical or asymmetrical modes?*

## References

[1] Matthew Durey and Paul A Milewski. "Resonant triad interactions of gravity waves in cylindrical basins". In: *Journal of Fluid Mechanics* 966 (2023), A25. ISSN: 0022-1120.

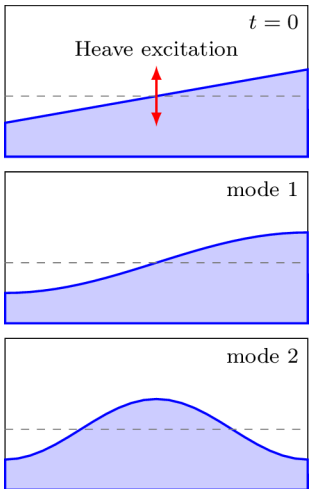



Figure 2.1: Faraday wave modes